

Code:

```
#include <stdio.h>

#define MAXN 300005

int parent_node[MAXN];
int weight_to_root[MAXN];

int find(int i) {
    if (parent_node[i] == i) return i;
    int root = find(parent_node[i]);
    weight_to_root[i] ^= weight_to_root[parent_node[i]];
    return parent_node[i] = root;
}

int dfs1(int np, int lst) {
    return np ^ lst;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;

    for (int i = 1; i <= n; i++) {
        parent_node[i] = i;
        weight_to_root[i] = 0;
    }

    while (q--) {
        int u, v, x;
        scanf("%d %d %d", &u, &v, &x);

        int root_u = find(u);
        int root_v = find(v);

        if (root_u != root_v) {
            parent_node[root_u] = root_v;
            weight_to_root[root_u] = weight_to_root[u] ^ weight_to_root[v] ^ x ^ 1;
            printf("YES\n");
        } else {
            if ((weight_to_root[u] ^ weight_to_root[v]) == (x ^ 1)) {
```

```
        printf("YES\n");
    } else {
        printf("NO\n");
    }
}
}
return 0;
}
```

Input:

```
9 12
6 10
1 31
3 60
6 20
6 41
3 41
2 40
2 50
4 50
7 81
8 91
9 70
```

Output:

```
9 12
6 10
1 31
3 60
6 20
6 41
```

3 4 1

2 4 0

2 5 0

4 5 0

7 8 1

8 9 1

9 7 0